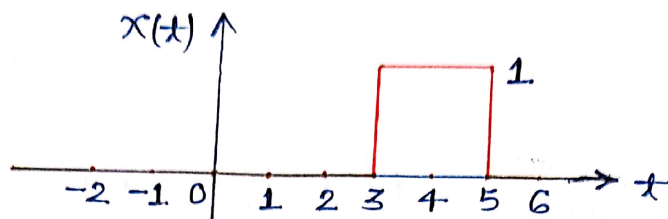
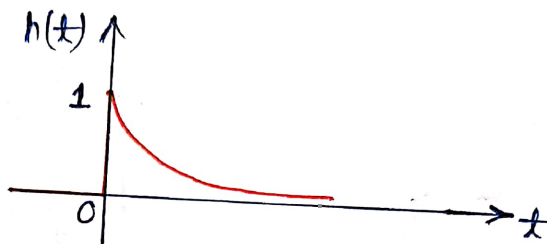


S&S QUIZ-3 SOLUTION

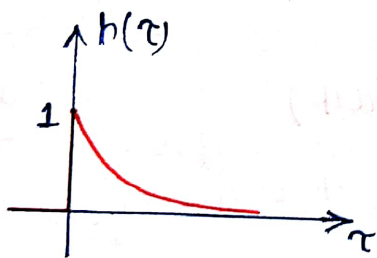
Sol(1)÷ Given- $x(t) = u(t-3) - u(t-5)$



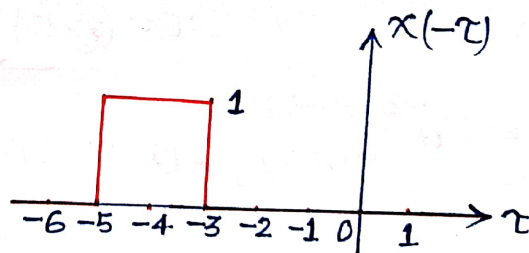
$$h(t) = e^{-3t} u(t)$$



Part(a)÷ $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) d\tau$
 $= \int_{-\infty}^{\infty} e^{-3\tau} u(\tau) \cdot \{u(t-\tau-3) - u(t-\tau-5)\} d\tau$ → (0.5 Mark)



→ (0.25 Mark)



→ (0.25 Mark)

Case(I)÷

For $t \leq 3$, $y(t) = 0$

→ (1 Mark)

Case(II)÷

For $3 < t \leq 5$, $y(t) = \int_0^{t-3} e^{-3\tau} d\tau = \left\{ \frac{e^{-3\tau}}{-3} \right\}_0^{t-3} = \left(\frac{1 - e^{-3(t-3)}}{3} \right)$

→ (1 Mark)



Case (III) ÷

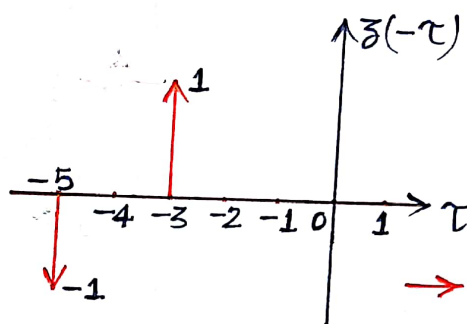
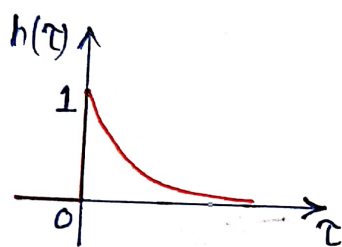
$$\text{For } t > 5, \quad y(t) = \int_{t-5}^{t-3} e^{-3\tau} d\tau = \left\{ \frac{e^{-3\tau}}{-3} \right\}_{t-5}^{t-3} = \left(\frac{e^{-3(t-5)}}{-3} - \frac{e^{-3(t-3)}}{-3} \right)$$

→ (1 Mark)

Part (b) ÷

$$g(t) = \left\{ \frac{dx(t)}{dt} \right\} * h(t) = \dot{x}(t) * h(t)$$

$$= \int_{-\infty}^{\infty} \dot{x}(\tau) \cdot h(t-\tau) d\tau = \int_{-\infty}^{\infty} h(\tau) \cdot \dot{x}(t-\tau) d\tau \rightarrow (0.5 \text{ Mark})$$



→ (0.5 Mark)

$$\therefore g(t) = \int_{-\infty}^{\infty} \left\{ e^{-3\tau} \cdot u(\tau) \right\} \left\{ -\delta(t-\tau-5) + \delta(t-\tau-3) \right\} d\tau$$

$$= - \int_{-\infty}^{\infty} e^{-3\tau} \delta(t-\tau-5) \cdot u(\tau) d\tau + \int_{-\infty}^{\infty} e^{-3\tau} \delta(t-\tau-3) u(\tau) d\tau$$

$$t-\tau-5=0$$

$$\tau = (t-5)$$

→ (0.5 Mark)

$$t-\tau-3=0$$

$$\tau = (t-3)$$

→ (0.5 Mark)

$$= -e^{-3(t-5)} \cdot u(t-5) \int_{-\infty}^{\infty} \delta(t-\tau-5) \cdot d\tau + e^{-3(t-3)} \cdot u(t-3) \int_{-\infty}^{\infty} \delta(t-\tau-3) d\tau$$

$$= -e^{-3(t-5)} u(t-5) + e^{-3(t-3)} u(t-3) \left\{ \begin{array}{l} \int_{-\infty}^{\infty} \delta(t-\tau-5) d\tau = 1 \\ \int_{-\infty}^{\infty} \delta(t-\tau-3) d\tau = 1 \end{array} \right\} \rightarrow (2 \times 0.5 \text{ Mark})$$

$$g(t) = e^{-3(t-3)} \cdot u(t-3) - e^{-3(t-5)} \cdot u(t-5)$$

→ (1 Mark)

Part (c):

As we know from Part (a),

$$y(t) = \begin{cases} 0, & t \leq 3 \\ \left(\frac{1 - e^{-3(t-3)}}{3} \right), & 3 < t \leq 5 \\ \left(\frac{e^{-3(t-5)} - e^{-3(t-3)}}{3} \right), & 5 < t \end{cases}$$

$$\frac{d}{dt} y(t) = \begin{cases} 0, & t \leq 3 \\ e^{-3(t-3)}, & 3 < t \leq 5 \\ e^{-3(t-3)} - e^{-3(t-5)}, & 5 < t \end{cases}$$

$$\begin{aligned} \therefore \frac{d}{dt} y(t) &= e^{-3(t-3)} \{ u(t-3) - u(t-5) \} \\ &\quad + \{ e^{-3(t-3)} - e^{-3(t-5)} \} u(t-5) \\ &= e^{-3(t-3)} \cdot u(t-3) - e^{-3(t-5)} \cdot u(t-5) \\ &= g(t) \end{aligned}$$

→ (2 Mark)

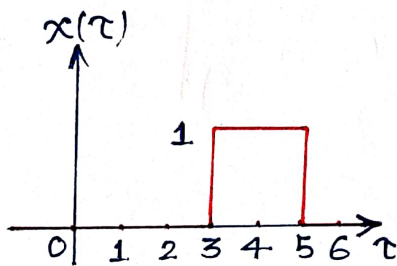
Hence, $g(t) = \frac{d}{dt} y(t)$

Alternate Method :-

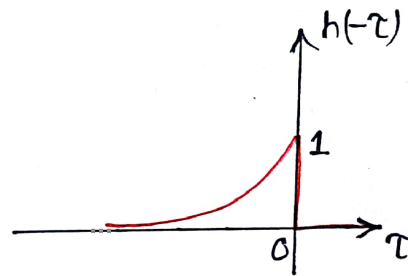
Sol 1(a) :-

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} \{u(\tau-3) - u(\tau-5)\} \cdot e^{-3(t-\tau)} \cdot u(t-\tau) d\tau \rightarrow (0.5 \text{ Mark})$$



$\rightarrow (0.25 \text{ Mark})$



$\rightarrow (0.25 \text{ Mark})$

Case(I) :- For $t \leq 3$, $y(t) = 0$

$\rightarrow (1 \text{ Mark})$

Case(II) :- For $3 < t \leq 5$, $y(t) = \int_3^t e^{-3(t-\tau)} d\tau = \left\{ \frac{e^{-3(t-\tau)}}{-3} \right\}_3^t$

$$= \left(\frac{1 - e^{-3(t-3)}}{-3} \right) \rightarrow (1 \text{ Mark})$$

Case(III) :- For $t > 5$, $y(t) = \int_3^5 e^{-3(t-\tau)} d\tau = \left\{ \frac{e^{-3(t-\tau)}}{-3} \right\}_3^5$

$$= \left(\frac{e^{-3(t-5)}}{-3} - \frac{e^{-3(t-3)}}{-3} \right) \rightarrow (1 \text{ Mark})$$